# Application Data Sheet 37 CFR 1.76

The Application Data Sheet is part of the provisional or non-provisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76.

## Inventor Information

# of inventors: 1

1. **Eric Haywood**

**Residence Information**
US Residency
Fort Collins, CO
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**Mailing Address**
149 Remington Street Unit 425
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## Application Information

| | |
|---|---|
| Customer number | **221639 -** |
| Correspondence address | --- |
| Title of invention | **LIQUID RUNTIME ARCHITECTURE WITH DYNAMIC MICROSERVICE MATERIALIZATION AND GPU PROPORTIONAL RESOURCE ALLOCATION** |
| Attorney docket number | **---** |
| Entity status | --- |

| | |
|---|---|
| Application type | Provisional Application under 35 USC 111(b) |
| Subject matter | Utility |
| Total number of drawing sheets | --- |
| Filing by reference | No |

## Representative Information

# of representatives: 0

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32).

Data was not provided for this section.

## Domestic Benefit/National Stage Information

# of benefit claims: 0

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c), 386(c), or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78(a)(2) or CFR 1.78(a)(4), and need not otherwise be made part of the specification.

Data was not provided for this section.

## Foreign Priority Information

# of foreign priority claims: 0

This section allows for the applicant to claim benefit of foreign priority and to identify any prior foreign application for which priority is not claimed. Providing this information in the Application Data Sheet constitutes the claim for priority as required by 35 U.S.C. 119 (b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX) the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a

copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Data was not provided for this section.

## Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

*Checking this box will cause the application to be examined under the first inventor to file provisions of the AIA.*

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013. NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 2016, 2013, will be examined under the first inventor to file provisions of the AIA.

## Authorization or Opt-Out of Authorization to Permit Access

When this Application Data Sheet is properly signed and filled with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

**NOTE:**
This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. **Priority Document Exchange (PDX)**
Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the China National Intellectual Property Administration (CNIPA), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its

related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h) (1).

B. **Search Results from U.S. Application to EPO**

Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with the search results from the instant application.

**NOTE:**

Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

## Applicant Information

# of applicants: 0

The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46.

Data was not provided for this section.

## Assignee Information including Non-Applicant Assignee Information

# of assignees: 1

An assignee-applicant identified in the "Applicant" section will appear on the patent application as an applicant.

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

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**Organization**

## Signature

**NOTE:**

This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). **However, if this Application Data Sheet is submitted with the INITIAL filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c)**

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See CFR 1.4(d) for the manner of making signatures and certifications.

| Signature | First name | Last name | Registration # | Date |
|---|---|---|---|---|
| /Eric Haywood/ | ERIC | HAYWOOD | --- | 03/03/2026 |

PROVISIONAL PATENT APPLICATION

Under 35 U.S.C. § 111(b)

# LIQUID RUNTIME ARCHITECTURE WITH DYNAMIC MICROSERVICE MATERIALIZATION AND GPU-PROPORTIONAL RESOURCE ALLOCATION

| | |
|---|---|
| Title of the Invention | Liquid Runtime Architecture with Dynamic Microservice Materialization and GPU-Proportional Resource Allocation |
| Docket Number | HS-2026-009 |
| Inventor(s) | Eric Haywood |
| Residence(s) | 149 Remington Street Unit 425, Fort Collins, CO 80524 |
| Correspondence Address | HeadySystems Inc., 149 Remington Street Unit 425, Fort Collins, CO 80524 |
| Attorney/Agent (Name & Reg. No.) | None |
| U.S. Government Interest (Yes/No) | No |
| Gov. Agency & Contract No. | N/A |

## Title of the Invention

[0001] Liquid Runtime Architecture with Dynamic Microservice Materialization and GPU-Proportional Resource Allocation

## Cross-Reference to Related Applications

[0002] This application claims the benefit of priority and is related to co-pending provisional applications filed concurrently herewith by the same inventor and assignee.

## Field of the Invention

[0003] This invention relates to systems and methods for dynamically materializing and dissolving cloud microservices based on real-time demand, with GPU resource allocation proportional to task computational intensity, and repository-aware projection reconciliation.

## Background of the Invention

[0004] Traditional cloud architectures require pre-provisioned microservices that consume resources whether in use or not. Container orchestration platforms provide scaling but cannot dynamically synthesize entirely new service types at runtime. No existing system materializes microservices from templates injected via a three-dimensional spatial workspace, allocates GPU resources proportionally using golden ratio scaling, or reconciles the live runtime state against its source repository projection to identify stray or untracked service artifacts.

## Summary of the Invention

[0005] The invention provides a system and method as described herein.

[0006] Dynamic Service Materialization: Cloud microservices are materialized on demand from capability templates stored in a 3D spatial workspace. Services exist only while needed and dissolve when their tasks complete, leaving no persistent infrastructure footprint.

[0007] Capability Registry: A dynamic registry of runtime capabilities (e.g., chat, music-ai, devtools, research, security, analytics) with health monitoring, version tracking and automatic failover.

[0008] GPU-Proportional Allocation: GPU resources are allocated proportionally to task computational intensity using golden-ratio-derived allocation thresholds, preventing both over-provisioning and starvation.

[0009] 3D Workspace Template Injection: Service templates are injected from a three-dimensional vector workspace into the runtime, carrying their full configuration, dependencies and scaling parameters as spatial coordinates.

[0010] Repository Projection Reconciliation: The runtime continuously compares its live state against the source repository, identifying untracked files, stray service artifacts and configuration drift, with optional automated reconciliation.

## Detailed Description of the Preferred Embodiment

A. LiquidUnifiedRuntime Architecture

[0011] The LiquidUnifiedRuntime orchestrates the lifecycle of cloud-only microservices. It maintains a registry of capabilities, each described by a name, sources, semantic version, health status and last health-check timestamp. Default capabilities include heady-chat, music-ai, devtools, research, security, memory, analytics, creative, operations, quality, finops and networking.

[0012] To optimize the service footprint, the runtime enforces a configurable maximum number of concurrent microservices. When this limit is exceeded, it scores services by last access, request volume, error rate and priority, then gracefully dissolves the lowest-scoring services by draining connections, persisting state and releasing resources.

B. GPU Allocation

[0013] GPU resources such as NVIDIA A100 and T4 instances are allocated proportionally to task intensity using golden-ratio thresholds. Intensity ranges of [0,0.382], [0.382,0.618], [0.618,0.854] and [0.854,1.0] correspond to φ-derived values ($\varphi^{-2} \approx 0.382$, $\varphi^{-1} \approx 0.618$, and $1 - \varphi^{-2} \approx 0.854$). Tasks in each range are assigned specific GPU types and counts, ensuring balanced utilization without over-provisioning.

[0014] The golden ratio ($\varphi \approx 1.618$) is the most irrational number and has the slowest-converging continued fraction. Splitting resources using φ-derived ratios produces self-similar partitions that minimize resonance patterns and contention, leading to efficient resource utilization.

C. 3D Workspace Injection

[0015] Capability templates are stored in a 3D spatial workspace where each dimension represents semantic domain, temporal state and hierarchy level. When a template is injected, the runtime extracts spatial coordinates, validates the template against the capability registry, materializes the corresponding microservice on the target cloud provider, registers it with health monitoring and returns the service endpoint.

[0016] Spatial indexing of templates may employ octree data structures, which partition three-dimensional space into eight child octants. Octrees enable O(log N) operations for insertion, deletion and search and are memory efficient because they store data only in leaf nodes.

D. Repository Projection Reconciliation

[0017] The runtime compares its live file system against the Git repository projection. It identifies untracked files, filters known generated files (such as node_modules, dist and coverage) and protected worker markers, reports stray files and optionally applies automated reconciliation by adding or removing files as needed.

## Claims

1. A computer-implemented method for managing cloud microservices, comprising: (a) dynamically materializing microservices from capability templates when demand is detected; (b) maintaining a capability registry with health monitoring, version tracking and automatic failover; (c) allocating GPU resources proportionally to task computational intensity using golden-ratio-derived allocation thresholds; (d) dissolving microservices when their tasks complete to maintain a configurable maximum concurrent service count; and (e) injecting service templates from a three-dimensional spatial workspace using coordinates representing semantic domain, temporal state and hierarchy level.

2. The method of claim 1, wherein the GPU allocation thresholds are derived from the golden ratio inverse ($\varphi^{-1} \approx 0.618$), golden ratio inverse squared ($\varphi^{-2} \approx 0.382$) and their complements, producing mathematically optimal allocation boundaries.

3. The method of claim 1, further comprising continuously reconciling the runtime's live state against a source repository projection by identifying untracked files, filtering protected artifacts, reporting configuration drift and optionally applying automated reconciliation.

4. A liquid runtime system comprising: a capability registry maintaining health-monitored cloud microservice descriptors; a materialization engine that instantiates services from 3D workspace templates on demand; a GPU allocation engine that assigns computational resources proportionally using golden-ratio scaling; a dissolution engine that gracefully removes underutilized services; and a reconciliation engine that compares live runtime state against a source repository projection.

## Implementation Reference

[0018] The Liquid Unified Runtime may be implemented in the file `src/services/liquid-unified-runtime.js` with dependencies such as `heady-principles.js` for phi-scaling functions. The runtime uses octree spatial indexing to manage capability templates in the 3D workspace.

## References

[1] Octrees partition three-dimensional space into eight child octants, enabling O(log N) operations for insertion, deletion and search and providing memory-efficient spatial indexing【135796677921791†L142-L156】

[2] Octrees store data only at leaf nodes and can prune empty branches, making them memory efficient for representing 3D space【135796677921791†L219-L229】

[3] The golden ratio has the slowest-converging continued fraction and is considered the most irrational number, making φ-derived splits ideal for resource allocation【577248341085003†L203-L218】

UNITED STATES
PATENT AND TRADEMARK OFFICE

P.O. Box 1450
Alexandria, VA 22313 - 1450
www.uspto.gov

# ELECTRONIC ACKNOWLEDGEMENT RECEIPT

| APPLICATION # | RECEIPT DATE / TIME | ATTORNEY DOCKET # |
|---|---|---|
| **63/995,325** | **03/03/2026 02:17:23 PM Z ET** | **-** |

## Title of Invention

LIQUID RUNTIME ARCHITECTURE WITH DYNAMIC MICROSERVICE MATERIALIZATION AND GPU PROPORTIONAL RESOURCE ALLOCATION

## Application Information

| | | | |
|---|---|---|---|
| APPLICATION TYPE | Utility - Provisional Application under 35 USC 111(b) | PATENT # | - |
| CONFIRMATION # | 1182 | FILED BY | ERIC HAYWOOD |
| PATENT CENTER # | 74697257 | FILING DATE | - |
| CUSTOMER # | 221639 | FIRST NAMED INVENTOR | Eric Haywood |
| CORRESPONDENCE ADDRESS | - | AUTHORIZED BY | - |

## Documents

**TOTAL DOCUMENTS: 2**

| DOCUMENT | PAGES | DESCRIPTION | SIZE (KB) |
|---|---|---|---|
| generatedADS74697257.pdf | 5 | Application Data Sheet | 109 KB |
| HS-2026-009-Final-Assigned-Updated-APP.TEXT.docx | 5 | Application body structured text document | 39 KB |

Warning: One or more pages are missing page numbering. Page numbering will be automatically applied after submission.

## Digest

| DOCUMENT | MESSAGE DIGEST(SHA-512) |
|---|---|
| generatedADS74697257.pdf | FB3250CF4E8799B0CBF3A10D4AFBFED2C668993C731143D1 |

7F85303AACBCB4B85B790E84EF46F3E531933B975AE55EA96
93E9F234068F632E35ED25F9126E8D4

HS-2026-009-Final-Assigned-Updated-APP.TEXT.docx

5B2A3E7A77BEE5DC9AD1A8F57E596ED145E1215423C8E029
73855DEC73C6976DF72934C3D64FCAD88090138428D30B058
A749B5D95A9350DC3F8CADCDCC38B9A

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

**New Applications Under 35 U.S.C. 111**

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

**National Stage of an International Application under 35 U.S.C. 371**

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

**New International Application Filed with the USPTO as a Receiving Office**

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.